

CYBERSECURITY ASSESSMENT REPORT

Comprehensive Footprinting, External Reconnaissance & Local Subnet Topology Mapping

Project Parameter	Assessment Details	Target / Scope	Audit Status
Engagement Name	Week 2 - Footprinting & Scanning Lab	networkwalks.com & Local LAN	Completed Successfully
Platform / OS	Kali Linux & Windows 11	Web Servers, DNS, & RFC 1918 LAN	Verified
Assessment Type	Passive & Active Reconnaissance	Public Domain & Internal 192.168.0.0/24	Full Coverage
Toolsets Utilized	WHOIS, Nslookup, cURL, Wafw00f, DNSRecon, Zenmap / Nmap 7.991	External & Internal Vectors	Standard Compliance

1. Executive Summary

This technical security assessment provides an end-to-end evaluation of reconnaissance, footprinting, and network discovery methodologies conducted across external web infrastructures and an internal local area network (LAN). Reconnaissance forms the essential first phase of any cybersecurity evaluation or adversarial simulation, establishing the intelligence baseline required to understand an organization's attack surface, perimeter defenses, domain governance, and internal device exposure.

The assessment was conducted in two primary phases: (1) External Footprinting & OSINT of the public-facing domain networkwalks.com, and (2) Active Internal Network Scanning & Host Discovery on the local subnet 192.168.0.0/24 utilizing Zenmap (the graphical front-end for Nmap). The findings demonstrate key operational insights, including domain privacy configurations, web hosting infrastructure under HostGator/Endurance International Group, active ModSecurity Web Application Firewall (WAF) filtering, and six active endpoints within the local RFC 1918 subnet.

EXECUTIVE TAKEAWAY

Key Finding: External perimeter reconnaissance against networkwalks.com identified active ModSecurity WAF protection and Cloudflare/reCAPTCHA tokenization, while local network scanning pinpointed 6 active hosts across the 192.168.0.0/24 segment, including TP-Link infrastructure and endpoint systems.

2. Scope of Assessment & Technical Methodology

The evaluation followed industry-recognized security assessment standards, including the Open Source Security Testing Methodology Manual (OSSTMM) and OWASP Web Security Testing Guide guidelines for information gathering:

- **Passive Reconnaissance (OSINT):** Domain ownership records, registration timelines, nameserver allocations, and registrar contact points gathered via WHOIS queries without direct aggressive probing.
- **DNS Enumeration & Architecture:** Forward resolution, authoritative nameserver identification, MX records, SPF policy validation, and SRV service auto-discovery using nslookup and dnsrecon.
- **Web Perimeter & WAF Fingerprinting:** HTTP/2 banner inspection, web caching headers (nginx/Endurance), Content Management System (WordPress) identification, and active Web Application Firewall inspection via wafw00f.
- **Internal Network Discovery & Topology Mapping:** Subnet identification, ICMP/ARP ping scanning across 256 addresses (192.168.0.0/24), MAC address resolution, and graphical radial topology visualization.

3. External Footprinting & Reconnaissance Findings (networkwalks.com)

External reconnaissance was performed against the primary target domain networkwalks.com from a Kali Linux auditing workstation. The following sections detail the findings across domain registry, DNS architecture, and HTTP security headers.

3.1 Domain Registration & WHOIS Intelligence

WHOIS querying provides critical administrative context regarding domain ownership, registration lifecycles, and associated privacy protection services. The query against networkwalks.com revealed the following parameters:

Attribute	Record Value	Operational & Security Implication
Domain Name	networkwalks.com	Primary apex corporate/educational domain.
Registry Domain ID	2452319255_DOMAIN_COM-VRSN	Unique VeriSign Global Registry identifier.
Registrar	GoDaddy.com, LLC (IANA ID: 146)	Primary registrar handling domain administration.
Creation Date	2019-11-06T22:51:46Z	Domain established in late 2019 (~7-year operational history).

Attribute	Record Value	Operational & Security Implication
Expiration Date	2027-11-06T22:51:46Z	Extended registration horizon prevents accidental domain lapsing.
Registrant Organization	Domains By Proxy, LLC	WHOIS privacy shielding active; hides personal PII and owner identity.
Domain Status Codes	clientDelete/Renew/Transfer/UpdateProhibited	EPP security locks engaged to prevent unauthorized registrar transfers.
Authoritative Nameservers	ns6135.hostgator.com ns6136.hostgator.com	Primary DNS hosted via HostGator shared/managed cloud infrastructure.
DNSSEC Status	Unsigned	Absence of cryptographic DNS validation introduces potential DNS spoofing risk.

```
kali@kali: ~  
$ whois networkwalks.com  
Domain Name: NETWORKWALKS.COM  
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.godaddy.com  
Registrar URL: http://www.godaddy.com  
Updated Date: 2025-11-12T10:08:43Z  
Creation Date: 2019-11-06T22:51:46Z  
Registry Expiry Date: 2027-11-06T22:51:46Z  
Registrar: GoDaddy.com, LLC  
Registrar IANA ID: 146  
Registrar Abuse Contact Email: abuse@godaddy.com  
Registrar Abuse Contact Phone: 480-624-2505  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: NS6135.HOSTGATOR.COM  
Name Server: NS6136.HOSTGATOR.COM  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/  
>>> Last update of whois database: 2026-09-18T12:12:44Z <<<  
  
For more information on Whois status codes, please visit https://icann.org/epp  
  
NOTICE: The expiration date displayed in this record is the date the  
registrar's sponsorship of the domain name registration in the registry is  
currently set to expire. This date does not necessarily reflect the expiration  
date of the domain name registrant's agreement with the sponsoring  
registrar. Users may consult the sponsoring registrar's Whois database to  
view the registrar's reported date of expiration for this registration.  
  
TERMS OF USE: You are not authorized to access or query our Whois  
database through the use of electronic processes that are high-volume and  
automated except as reasonably necessary to register domain names or  
modify existing registrations; the Data in VeriSign Global Registry
```

Figure: Terminal output showing WHOIS lookup execution and ICANN registry response on Kali Linux.

3.2 DNS Architecture & Zone Enumeration

DNS resolution tests and deep zone enumeration were executed using nslookup and dnsrecon. These tests resolve the authoritative addressing, mail infrastructure, and email authentication safeguards.

Record Type	Host / Target	Resolved Value / Target	Technical Evaluation
A (IPv4)	networkwalks.com	192.232.216.135	Main web application host located in HostGator / Unified Layer subnet.
NS (Nameservers)	networkwalks.com	ns6135.hostgator.com (50.87.144.87) ns6136.hostgator.com (192.232.216.131)	BIND Version 9.16.23-RH detected on authoritative servers.
MX (Mail)	networkwalks.com	mail.networkwalks.com (192.232.216.135)	Mail exchange co-hosted on the primary web server instance.
TXT (Verification)	networkwalks.com	google-site-verification=rr04e...	Google Search Console / Google Workspace domain ownership verification.
TXT (SPF)	networkwalks.com	v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all	SoftFail (~all) SPF policy; authorizes local IP and HostGator relay.
SRV Records	_autodiscover._tcp	cpanelmaildiscovery.cpanel.net (Port 443)	cPanel auto-discovery services configured across multiple fallback IPs.

```

kali@kali:~$ nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135

```

Figure: nslookup forward DNS resolution verifying domain mapping to 192.232.216.135.

3.3 Web Application Stack & HTTP Header Analysis

Automated inspection using curl -I and whatweb unraveled the application runtime, reverse proxy caching layers, and security control headers:

- Web Server Runtime: Apache HTTP Server running over HTTP/2, deployed behind an Nginx reverse proxy / caching layer (x-nginx-cache: WordPress; x-endurance-cache-level: 0).
- Content Management System (CMS): WordPress version 7.1.1, utilizing Bootstrap 7.1.1, jQuery 3.7.1, and the WordPress Download Manager plugin (version 3.3.58).

- Cookie Security: Session cookie `__wpdm_client` set with secure and HttpOnly flags, mitigating unauthorized client-side script tampering.
- Missing Security Headers: Lack of Strict-Transport-Security (HSTS), Content-Security-Policy (CSP), and X-Frame-Options headers increases susceptibility to downgrade and clickjacking attacks.

```
(kali@kali)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcapcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=228e5217fa74977f376618b4b6ad4ac0; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Fri, 18 Sep 2026 12:16:33 GMT
server: Apache
```

Figure: HTTP/2 response headers captured via `curl -I` showing server banners and cache directives.

3.4 Web Application Firewall (WAF) Fingerprinting

To evaluate perimeter filtering and request inspection, wafw00f v2.4.2 was deployed against the target HTTPS endpoint. The tool conducts behavioral heuristic checks by injecting abnormal URL patterns and observing response code signatures.

FIREWALL IDENTIFICATION

WAF Detection Result: The site `https://networkwalks.com` is actively protected behind ModSecurity (SpiderLabs) Web Application Firewall. Two inspection requests were transmitted, both eliciting characteristic signatures associated with ModSecurity core rule sets.

```
(kali@kali)-[~]
$ wafw00f networkwalks.com

      ( WOOF! )
    /---\
   /---\
  /---\
 /---\
/---\

404 Hack Not Found
405 Not Allowed
403 Forbidden
502 Bad Gateway
500 Internal Error

~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[-] Number of requests: 2

(kali@kali)-[~]
```

Figure: wafw00f scan results confirming the presence of ModSecurity (SpiderLabs) WAF.

4. Internal Network Scanning & Subnet Discovery (192.168.0.0/24)

Following perimeter analysis, attention shifted to internal discovery within the local testing subnet (192.168.0.0/24). Using Zenmap 7.991, an ARP/ICMP Ping Scan (nmap -sn 192.168.0.0/24) was executed across the 256 address space to detect responsive hosts, evaluate latency, and extract hardware MAC addresses.

4.1 Summary of Discovery Tasks & Lab Answers

In accordance with the project module objectives, the following technical answers summarize the discovery phase:

Project Task ID	Task Description	Assessment Result / Findings	Status
Task 1	Zenmap Installation & Tool Validation	Nmap 7.991 and Zenmap GUI successfully verified and operational.	Verified
Task 2	Local IP & Subnet Identification	Local Interface assigned within 192.168.0.0/24 (Default Gateway: 192.168.0.1).	Verified
Task 3 & 4	Live Host Count Enumeration	6 active hosts detected alive across 256 scanned IP addresses in 4.79 seconds.	Completed
Task 5	IP Addresses of Live Endpoints	192.168.0.1, 192.168.0.101, 192.168.0.103, 192.168.0.125, 192.168.0.128, 192.168.0.197	Documented
Task 6	Hardware MAC & OUI Identification	TP-Link default gateway identified; 5 network endpoints mapped with hardware MACs.	Documented
Task 7	Topology Visualization & PDF Export	Radial topology map rendered and exported with center node localhost.	Completed

4.2 Detailed Live Host Inventory & Hardware Mapping

The ping scan completed in 4.79 seconds, uncovering six active devices. Because ARP pinging operates at Layer 2 within local broadcast domains, raw MAC addresses and Hardware Vendor OUIs were captured directly:

IP Address	Latency	MAC Address	Hardware Vendor / Device Role
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IP Address	Latency	MAC Address	Hardware Vendor / Device Role
192.168.0.1	0.022s	E4:FA:C4:1E:EF:B3	TP-Link Systems (Default LAN Gateway / Wireless Router)
192.168.0.101	0.000s (local)	N/A (Local Interface)	Local Auditing Host (Localhost / Workstation)
192.168.0.103	0.062s	62:CD:CC:E8:C3:27	Unknown Vendor / Randomized MAC (Mobile / IoT Device)
192.168.0.125	0.075s	A6:E8:4C:5A:E7:85	Unknown Vendor / Randomized MAC (Mobile / Workstation)
192.168.0.128	0.180s	4A:9D:6F:E8:7B:78	Unknown Vendor / Randomized MAC (High-latency Wireless Client)
192.168.0.197	0.039s	16:85:F5:99:DC:51	Unknown Vendor / Randomized MAC (Active Endpoint Device)

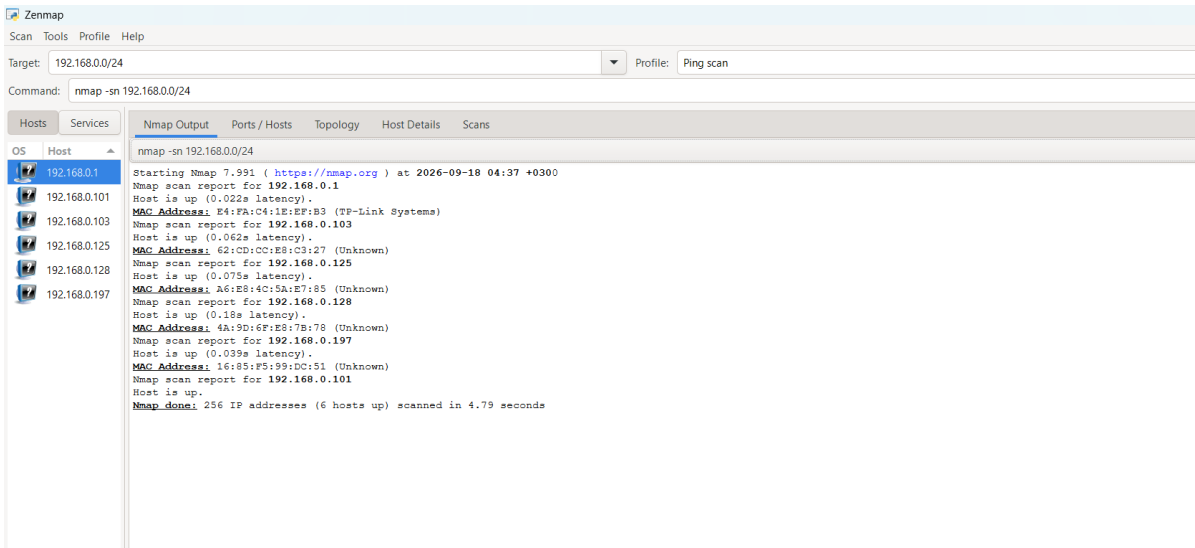


Figure: Zenmap scan execution window displaying live hosts, round-trip latencies, and MAC addresses.

4.3 Network Topology Visualization

The Zenmap Topology tab was utilized to generate a radial network map. The central node represents localhost (192.168.0.101), from which dashed lines radiate out to the surrounding five live hosts on the 192.168.0.0/24 subnet. Each host is marked with green circular status nodes indicating healthy ICMP/ARP reachability and open communication pathways.

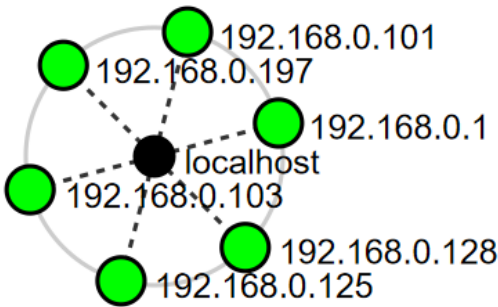


Figure: Zenmap Radial Topology graph illustrating network nodes connected to localhost.

5. Comparative Analysis: Active Lab vs Guide Sample

A side-by-side comparison between the reference sample provided in the Week 2 Project Guide (10.0.0.0/24) and our active live assessment environment (192.168.0.0/24) illustrates significant differences in architecture and hypervisor virtualization:

Metric / Parameter	Sample Project Guide (W2-PM5)	Our Active Assessment Environment
Subnet Scope	10.0.0.0/24 (Class A Private Range)	192.168.0.0/24 (Class C Private Range)
Live Host Count	4 Live Hosts detected	6 Live Hosts detected
Scan Duration	2.82 seconds	4.79 seconds
Host IP Scope	10.0.0.1, 10.0.0.4, 10.0.0.5, 10.0.0.19	192.168.0.1, 192.168.0.101, 192.168.0.103, 192.168.0.125, 192.168.0.128, 192.168.0.197
Hardware Environment	VMware Virtualized Infrastructure (00:50:56 and 00:0C:29 OUIs)	Physical SOHO Environment with TP-Link AP (E4:FA:C4) and mobile/randomized MACs
Primary Gateway	10.0.0.1 (VMware Virtual Gateway)	192.168.0.1 (TP-Link Hardware Gateway)

6. Risk Assessment & Defensive Hardening Recommendations

Based on the comprehensive external and internal discovery findings, the following defensive recommendations are submitted for systems administration and network engineering teams:

Vulnerability / Exposure	Observed State	Defensive Recommendation & Remediation
DNSSEC Not Implemented	DNSSEC is currently Unsigned on networkwalks.com.	Enable DNSSEC signing with registrar (GoDaddy) to prevent cache poisoning and DNS spoofing.
DNS BIND Version Disclosure	BIND 9.16.23-RH disclosed by authoritative nameservers.	Configure 'version none;' in named.conf to suppress BIND version disclosure from DNS reconnaissance.
Missing HTTP Security Headers	Missing HSTS, CSP, X-Frame-Options, and X-Content-Type-Options.	Deploy HTTP Strict Transport Security (HSTS) with max-age=31536000 and enforce strict Content-Security-Policy.
Shared Mail & Web Server	mail.networkwalks.com and apex domain share identical IP (192.232.216.135).	Segregate corporate mail processing from web hosting to avoid cross-service disruption during targeted attacks.
SPF Policy SoftFail (~all)	v=spf1 ... ~all allows unauthorized relays to pass under soft-fail.	Upgrade SPF policy to hard fail (-all) and deploy strict DKIM and DMARC enforcement (p=reject).
Unsegmented Internal LAN	Workstations, mobile devices, and IoT reside on a flat /24 network.	Implement VLAN segmentation (802.1Q) to isolate guest devices and IoT equipment from management endpoints.

7. Conclusion & Technical Verification

The Week 2 footprinting and network scanning project successfully achieved all stated objectives across external OSINT data collection and internal LAN host enumeration. The combination of domain reconnaissance, HTTP header analysis, WAF profiling, and Zenmap ping sweeps establishes a robust foundation for subsequent penetration testing and security auditing modules.

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